

✓ Machine learning: Open problem in churn prediction

In the telecommunications business, "churn" refers to the movement of customers from one service provider to another. Service providers want "low churn" -- they want their customers to stick with them.

Can a service provider predict who will discontinue service based on factors like whether they are a senior, how long they have been a customer, whether they have internet security software, and how much they pay each month? Obviously, a service provider would like to be able to predict if a customer has a high probability of leaving.

You will complete this notebook by using the methods we have covered in class to try to predict the 'Churn' variable, which is a binary variable.

The dataset has about 7,000 rows, with each row representing a customer. The variables in the dataset tell us things like how long the customer has been with the service provider, what kind of contract they have, and whether the customer is a "senior citizen".

✓ Instructions:

- Use two machine learning algorithms to predict the value of the target variable 'Churn'.
- You will need to preprocess the data to prepare it for machine learning.
- You are expected to use the methods and best practice we've covered in class.
- The use of AI like ChatGPT for explicit solution is not allowed in this assignment, but you can use AI to help you understand task and give suggestions.
- You do not need to provide lots of text, but provide clear, brief text to explain what you are doing.
- You will need to add imports to the imports code cell below.
- Run your notebook from top to bottom before submitting. Cell execution numbers must be visible.

Refer to class materials to remember our recommended best practice for:

- data scaling
- test sets and cross validation
- assessing your prediction results
- hyperparameter tuning
- encoding categorical variables

For this notebook, you don't need to use or worry about:

- data visualization, except as you need for identifying variable types
- missing data

Tidiness of code and text is important, though. Please separate your notebook into sections using section headings. Use '### ' to indicate subheadings, as in this cell.

```
# Some commonly-used packages are imported below
import warnings
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split, cross_val_score, G
from sklearn.preprocessing import StandardScaler
```

```
# this is in preparation for Pandas 3.0
pd.options.mode.copy_on_write = True
```

```
# plotting
sns.set_theme(style='whitegrid', context='notebook')
plt.rcParams['figure.figsize'] = 5,3
```

```
# suppress Seaborn future warnings
warnings.simplefilter(action='ignore', category=FutureWarning)
```

✓ Read the data

```
df = pd.read_csv("https://raw.githubusercontent.com/grbruns/cst383/refs/
```

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 8 columns):
#   Column                Non-Null Count  Dtype
---  -
0   OnlineSecurity         7043 non-null  object
1   DeviceProtection       7043 non-null  object
2   PaperlessBilling       7043 non-null  object
3   SeniorCitizen          7043 non-null  int64
4   tenure                 7043 non-null  int64
5   MonthlyCharges         7043 non-null  float64
6   Contract               7043 non-null  object
7   Churn                  7043 non-null  object
```

```
dtypes: float64(1), int64(2), object(5)  
memory usage: 440.3+ KB
```

```
df.describe()
```

	SeniorCitizen	tenure	MonthlyCharges	
count	7043.000000	7043.000000	7043.000000	
mean	0.162147	32.371149	64.761692	
std	0.368612	24.559481	30.090047	
min	0.000000	0.000000	18.250000	
25%	0.000000	9.000000	35.500000	
50%	0.000000	29.000000	70.350000	
75%	0.000000	55.000000	89.850000	
max	1.000000	72.000000	118.750000	